Response to Reviewers — Second Cut

Team 3 · *Servo Characterization for the SOFIA Modular Kinetic Surface*

Acknowledgements

The authors thank the reviewers for their careful evaluation of the first-cut manuscript. <One sentence on how the comments improved the paper.> All comments were considered; the responses below follow the keys of the review report (R1.1, R1.2, . . .). Line, section, figure, table, and equation numbers refer to the revised manuscript that accompanies this letter.

Main changes. (i) <Main change 1> (Section I). (ii) <Main change 2>. (iii) <Main change 3>.

Summary of changes (observation–response–change matrix)

8 observations: 8 addressed, 0 partially addressed, 0 not addressed (justified).

Key	Status	Location in the revised paper
R2.4	**[Addressed]**	lines 36-47
R3.1	**[Addressed]**	Fig. 1
R3.3	**[Addressed]**	lines 193-199, Table VI, (1)
R3.5	**[Addressed]**	Table IV, Table V,Table VI
R3.6	**[Addressed]**	Table VIII
R3.8	**[Addressed]**	Table 1
R4.1	**[Addressed]**	lines 193-199, Table VI, (1)
R4.2	**[Addressed]**	Section I

Reviewer 1 — *Format, LaTeX and template compliance*

Reviewer 2 — *Writing and structure*

R2.4 [Addressed]

Reviewer comment: Salvo un párrafo final, Materials and Methods está formada solo por tablas; las subsecciones A–F no tienen texto que explique qué muestra cada tabla ni qué se concluye.

Response: We rewrote the introduction and all mentions of servo characterization to explicitly state that it is the proposed contribution of this work. We simplified the work to only have the servo characterization as our only differentiation. (lines 36-47). The revised text now reads:

> *"This work proposes an experimental method to estimate individual servo time constraints with a mechanical solution"*

Changes in the revised paper: lines 36-47

Reviewer 3 — *Technical content*

R3.1 [Addressed]

Reviewer comment: No hay arquitectura preliminar del sistema en figura: cómo se conectan la cámara y el PC, la ESP32-C3, los dos PCA9685, los 32 servos, los transceptores RS-485 y la alimentación

Response: Added a System Architecture Diagram Fig. 1 showing the host, ESP32-C3, dual PCA9685 drivers, 32 servos, RS-485 transceivers.

Changes in the revised paper: Fig. 1

R3.3 [Addressed]

Reviewer comment: La "constante de máquina" o "mechanical time constant" del servo no se define: no hay modelo, variables ni ecuaciones. El diferenciador promete compararla contra el "valor teórico" del MG90S, que la hoja de datos no reporta.

Response: Formulated the first-order dynamic model $\theta(t) = \theta_f + (\theta_0 - \theta_f)e^{-t/\tau_m}$ (1) and clarified that the reference baseline is derived from the manufacturer's no-load speed specification (0.10 s/60° at 4.8 V) Table VI.

Changes in the revised paper: lines 193-199, Table VI, (1)

R3.5 [Addressed]

Reviewer comment: Las fichas técnicas tienen solo parámetro, valor y verificación. No indican la versión ni la página de la hoja de datos, ni justifican el parámetro frente al requerimiento. El par del MG90S aparece como "1.8–2.2 kgf cm" sin decir a qué voltaje corresponde cada valor.

Response: Added source and justifications to the datasheets and updated the MG90S table to specify torque values at both 4.8V (1.8 kgf · cm) and 6.0V (2.2 kgf · cm).

Changes in the revised paper: Table IV, Table V,Table VI

R3.6 [Addressed]

Reviewer comment: El BOM no distingue la cantidad por unidad de la del sistema. Lista 1 ESP32-C3 y 2 PCA9685 pero 16 transceptores RS-485, sin explicar la arquitectura que eso implica. Usa "3D Printed" como fabricante, contiene elementos poco claros ("Pro3" de Tower Pro, "Axis 1 9mm x 9mm", "Inser M3x5x4") y lista conectores JST-PH, cuando la arquitectura de SOFIA usa JST-VH. Tampoco tiene descripción técnica ni localidad.

Response: Updated the BOM table to distinguish unit vs. system quantities, add descriptions for the parts, set the manufacturer to "In-house (LabDST, ITESO)", and added manufacturing origin.

Changes in the revised paper: Table VIII

R3.8 [Addressed]

Reviewer comment: El algoritmo usa r0 para la posición del módulo y ^v para la dirección, en lugar de la notación del curso (r0,2, ^r2,p). Además escribe atan2, asin y Clamp en itálica matemática, sin operatorname ni Call.

Response: Updated math operators to standard notation (atan2, arcsin, Clamp) and explicitly labeled the trajectory step as an Exponential Moving Average (EMA)

Changes in the revised paper: Table 1

Reviewer 4 — *D-Lab and state of the art*

R4.1 [Addressed]

Reviewer comment: El diferenciador propone asignación automática de direcciones Modbus y caracterización de la constante de máquina de cada servo. El artículo menciona ambas en la introducción, pero se enfoca en justificar la arquitectura electrónica (Línea 83) y no desarrolla ninguna de las dos.

Response: Formulated the first-order dynamic model $\theta(t) = \theta_f + (\theta_0 - \theta_f)e^{-t/\tau_m}$ (1) and clarified that the reference baseline is derived from the manufacturer's no-load speed specification (0.10 s/60° at 4.8 V) Table VI. We simplified the article to only use the servo mechanical time constant as differentiation

Changes in the revised paper: lines 193-199, Table VI, (1)

R4.2 [Addressed]

Reviewer comment: No hay planteamiento formal del problema, pregunta de investigación ni objetivos.

Response: We added a formal Problem Statement, Research Question, and Objectives in natural narrative format at the end of the Introduction (Section I), addressing per-servo time-constant estimation and actuation synchronization. The revised text now reads:

Changes in the revised paper: Section I

"Standard low-cost micro-servos exhibit manufacturing tolerances that result in non-uniform actuation speeds and transient response times across different units. In a 16-unit, 32-actuator modular kinetic surface like SOFIA, driving all servos with uniform control signals creates dynamic motion desynchronization and mechanical stress during normal use. To address this limitation, this work investigates the following research question: Can experimental estimation of individual servo mechanical time constants (τ_m) and their integration into a trajectory controller improve dynamic positional synchronization across a 32-actuator kinetic surface? This encompasses three primary objectives: (1) designing a PCB integrating an ESP32-C3 microcontroller, dual PCA9685 drivers, and RS-485 interfaces; (2) developing an automated micro-test protocol to capture step-response and extract individual time constants; and (3) integrating individual compensation parameters into the surface tracking controller to equalize dynamic settling times across all actuators."